IoT-Enabled Embedded LPG Safety & Emergency Response System

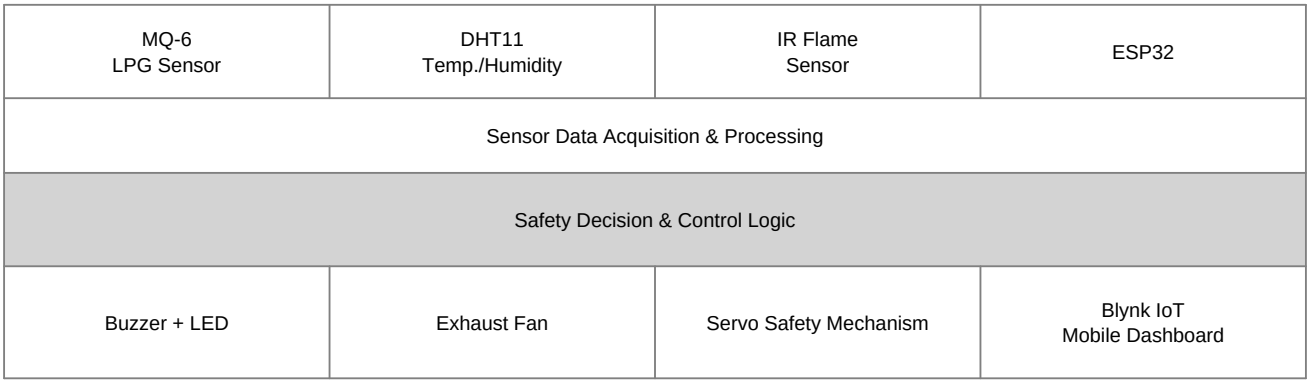
1. Problem Statement

LPG leakage in homes and small establishments can cause fire, explosion, health hazards and property damage. Conventional safety systems may only provide a local alarm and may not support continuous remote monitoring or automatic emergency response. A smart, low-cost system is therefore needed to detect hazardous conditions early, alert users remotely and initiate appropriate safety actions.

2. Methodology

The proposed system uses an ESP32 as the main controller. LPG leakage is detected using an MQ-6 gas sensor, while temperature and humidity are monitored using a DHT11 sensor and flame conditions are detected using an IR flame sensor. The ESP32 continuously processes these sensor readings and classifies the condition as normal or hazardous. During a detected hazard, the system activates a buzzer and LED indicators and can automatically operate an exhaust fan and a servo-based safety mechanism for demonstration. An OLED can display the real-time system status. The Blynk IoT platform provides remote monitoring, control and notifications through a smartphone. The prototype is an educational safety model and is not connected to or used to modify a real LPG cylinder or regulator.

3. Proposed Block Diagram



4. Proposed Outcome

The expected outcome is a compact working prototype capable of real-time LPG leakage and environmental monitoring, local emergency warning, automatic response demonstration and IoT-based remote supervision. The system aims to reduce response time during a suspected leakage event and demonstrate how embedded intelligence and IoT connectivity can improve LPG safety. The modular design can be extended with additional sensors and safety functions in future.

Key Technologies: ESP32 • MQ-6 LPG Sensor • DHT11 • IR Flame Sensor • OLED • Buzzer/LED • Exhaust Fan • Servo Mechanism • Blynk IoT